Bragi: quantum computing calibration toolkit at Chalmers

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Aim

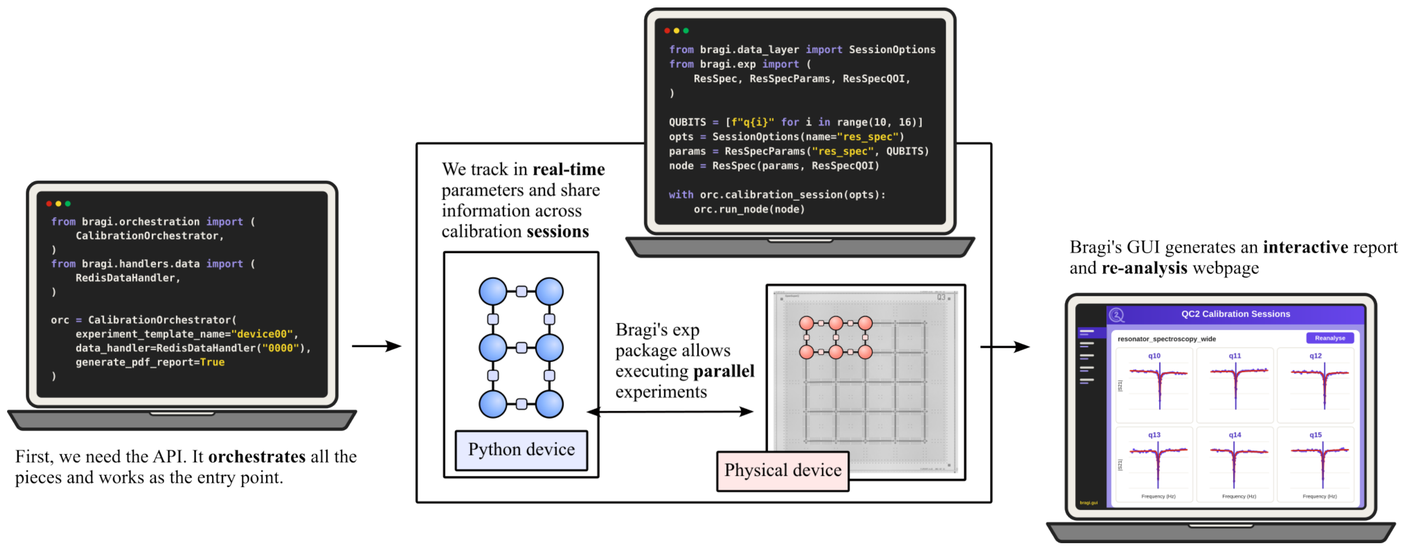
We present here our characterization and calibration toolkit, **Bragi**, for self-produced superconducting quantum processors, designed around Qblox electronics. Our lab design and produces multiple generations of devices, and this requires a flexible and scalable software stack to keep them operational. Among our goals are:

- **Modularity:** effortless use on different devices
- **Performance:** parallelization and speed
- **Reliability:** real-time calibration triage
- **Philosophy:** academic-driven and open-source

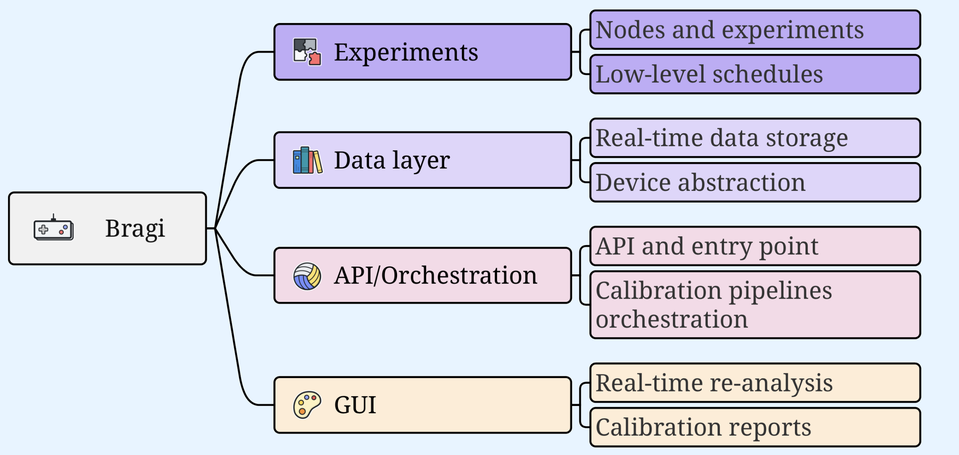
Software quality

- **Installation:** one command with `uv`, one linked workspace
- **Code quality:** `pre-commit` and CI with black, ruff, isort, ESLint, type and license-header checks
- **Efficient CI:** package tests only on changed code, full integration suite on every push
- **Testing:** 84% test coverage
- **Documentation:** fully documented with Sphinx (scan the QR code)

Overview of the workflow

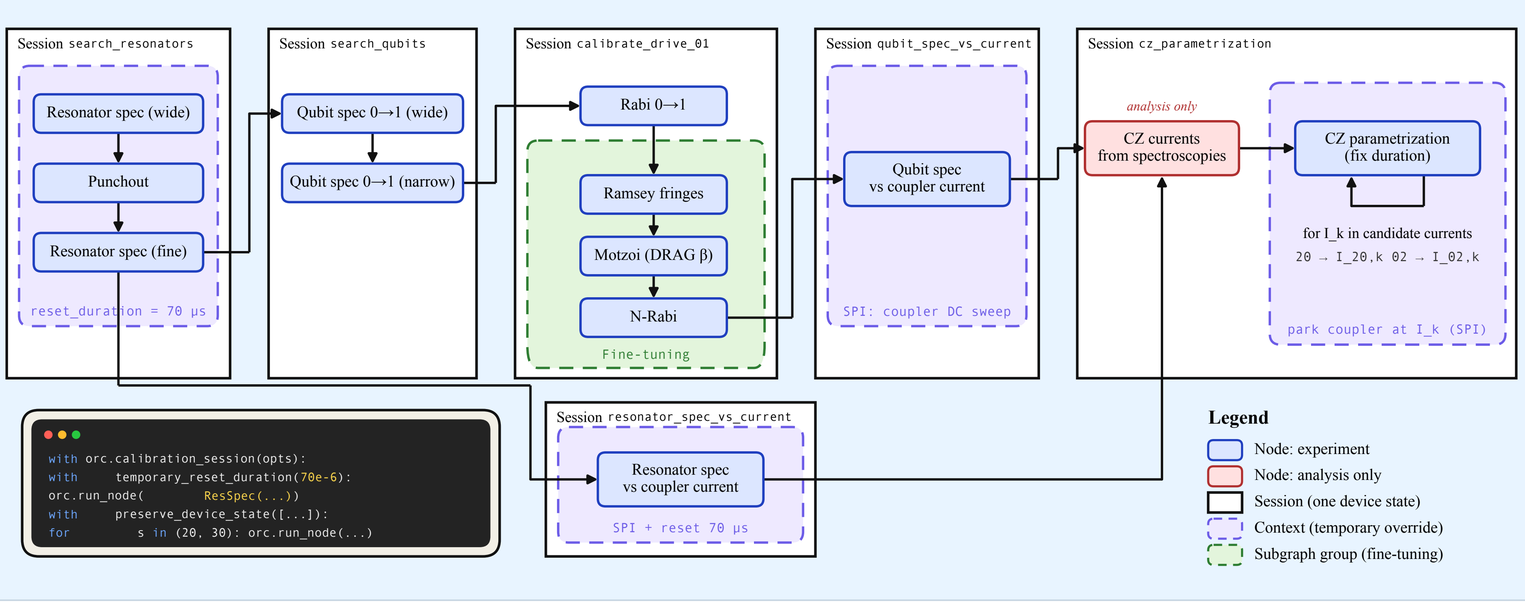


Four packages, one monorepo



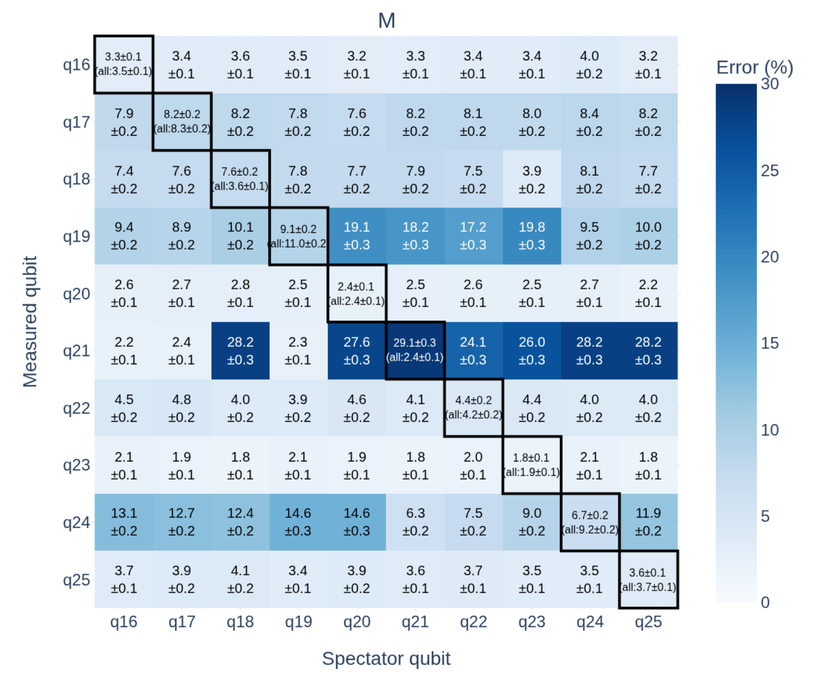
Complex sequences: CZ calibration

A calibration pipeline is not only a sequence of experiments: *i)* calibration sessions can group multiple experiments under some common settings, *ii)* logic may be required to implement loops and conditionals, and *iii)* sub-graphs can be embedded into higher-level optimization workflows.



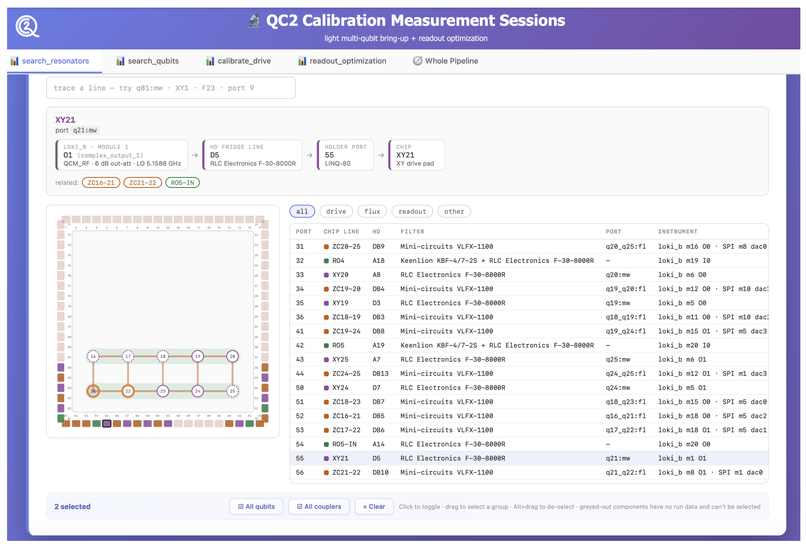
Physics application: readout crosstalk

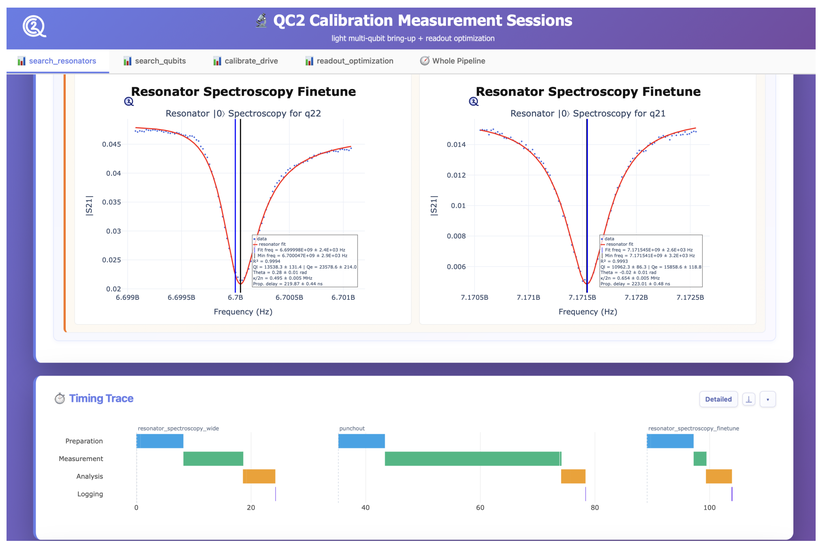
Complex physics experiments can be run with Bragi. Here, we show a slice of a readout crosstalk characterization.

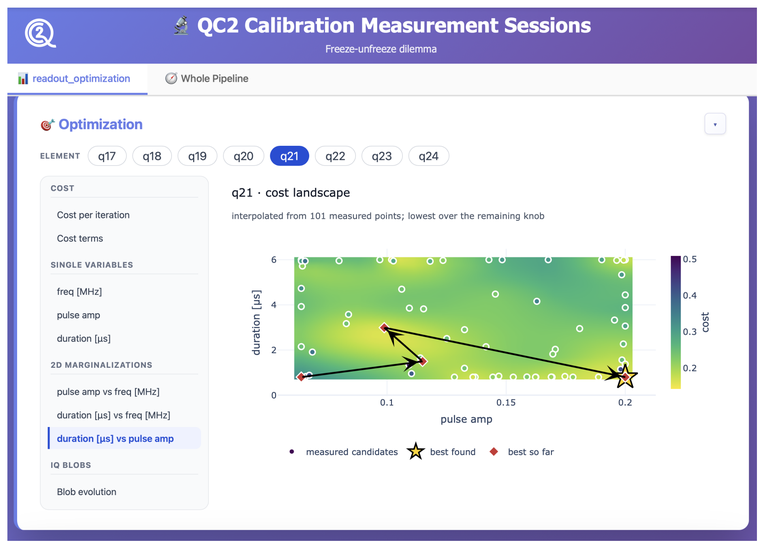


Visualization

Scaling to many qubits means we need smart ways to understand what is happening. Bragi's GUI turns every session into an interactive report: navigate the graph of nodes and their results, analyze time profiling, inspect chip's connections, and **re-analyse** them without measuring again.







Outlook

- **Open source:** releasing the code
- **Gates:** larger set of native gates and feedback capabilities
- **Error correction:** integration with a real-time QEC decoder
- **TWPA:** new calibration routines for TWPAs
- **Vendors:** support for additional hardware vendors
- **Scale:** monitoring and triage for systems of 100+ qubits
- **Quantum OS:** deeper interaction between the tools, towards a quantum operating system

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